

OPERATORS

Arithmetic Operators

These are used to perform basic math.

+ (Addition): Adds two values together.

$$5 + 3 = 8.$$

$$10 + 20 = 30.$$

- (Subtraction): Subtracts one value from another.

$$10 - 4 = 6.$$

$$100 - 25 = 75.$$

* (Multiplication): Multiplies two numbers.

$$4 * 3 = 12.$$

$$5 * 6 = 30.$$

/ (Division): Divides one number by another and always gives a decimal result.

$$12 / 4 = 3.0.$$

$$5 / 2 = 2.5.$$

// (Floor Division): Divides two numbers but rounds down to the nearest whole number.

$$\text{eg: } 5 // 2 = 2.$$

$$11 // 3 = 3.$$

% (Modulus): Gives you the leftover remainder after dividing.

$$9 \% 2 = 1$$

$$15 \% 4 = 3$$

** (Exponentiation): Raises the first number to the power of the second number.

$$2 ** 3 = 8 \ (2 \times 2 \times 2).$$

$$5 ** 2 = 25 \ (5 \times 5).$$

Assignment Operators

These are used to assign (save) values to variables.

= (Assignment): Assigns the value on the right to the variable on the left.

x = 5 (The variable x now holds the number 5).

: color = "blue" (The variable color now holds the text "blue").

+= (Add and Assign): Adds a value to a variable and saves the new result back to that variable.

If score = 10, doing score += 5 makes score become 15.

If age = 20, doing age += 1 makes age become 21.

-= (Subtract and Assign): Subtracts a value from a variable and saves the new result.

If health = 100, doing health -= 20 makes health become 80.

If items = 5, doing items -= 1 makes items become 4.

Comparison Operators

These compare two values and return either True or False.

== (Equal to): Checks if two values are exactly the same.

5 == 5 is True.

5 == 9 is False.

!= (Not Equal to): Checks if two values are different.

5 != 9 is True.

4 != 4 is False.

> (Greater than): Checks if the left value is bigger than the right value.

10 > 5 is True.

3 > 8 is False.

< (Less than): Checks if the left value is smaller than the right value.

4 < 9 is True.

12 < 5 is False.

Logical Operators

These are used to combine multiple conditions together.

and: Returns True only if both sides are true.

$(5 > 3)$ and $(10 > 2)$ is True (because both sides are true).

$(5 > 3)$ and $(2 > 10)$ is False (because the second part is false).

or: Returns True if at least one side is true.

$(5 > 3)$ or $(2 > 10)$ is True (because the first part is true).

$(1 > 5)$ or $(2 > 10)$ is False (because both parts are false).

not: Reverses the answer (turns True to False, and False to True).

$\text{not}(5 > 3)$ is False (because $5 > 3$ is true, and not flips it).

$\text{not}(1 > 5)$ is True (because $1 > 5$ is false, and not flips it).